

DATA 505 (Statistics Using R): Final Exam

Part 1: Closed Computer

Name: _____

2025-12-09

Instructions:

The exam will have two parts:

- **Part 1 (closed computer):** Multiple choice and short answer questions. You may not use your computer or any notes. You will write answers on this exam sheet.
- **Part 2 (open computer):** Coding and analysis problems. You may use your computer and any online resources that are "not alive" (i.e., you may freely use online resources, but you may not communicate with any other person). You will submit a completed Quarto document.
 - Once you turn in Part 1, you may open your computer and download the blank Quarto script for Part 2.
 - You may not return to Part 1 after turning it in and beginning the open-computer portion.

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TURN OVER WHEN INSTRUCTED

Part 1A (closed computer): Multiple choice

1 (3 points)

What is the result of evaluating the following R expression?

```
x <- 5
y <- x + 3
x <- 10
y
```

- A. 5
- B. 8
- C. 10
- D. 13

2 (3 points)

You have a data frame `students` with columns `name`, `age`, and `gpa`. Which R command correctly subsets the rows where GPA is greater than 3.5?

- A. `students[students$gpa > 3.5]`
- B. `students[students$gpa > 3.5, ]`
- C. `students[gpa > 3.5, ]`
- D. `filter(students > 3.5)`

3 (3 points)

A 95% confidence interval for the mean height of adult males in a population is (68 inches, 72 inches). Which statement is correct?

- A. 95% of males in the population have heights between 68 and 72 inches
- B. There is a 95% probability that the true population mean is between 68 and 72 inches
- C. If we repeated the sampling process many times, approximately 95% of the constructed intervals would contain the true population mean
- D. The sample mean must be 70 inches

4 (3 points)

A researcher wants to test whether the proportion of voters supporting a policy differs between urban and rural areas. Which statistical test is most appropriate?

- A. One-sample t-test
- B. Two-sample proportion test
- C. Paired t-test
- D. Simple linear regression

5 (3 points)

In a regression model predicting house price (in thousands) from square footage, the intercept is $\hat{\beta}_0 = 50$. What does this value represent?

- A. The average house price in the dataset
- B. The predicted price for a house with 50 square feet
- C. The predicted price (in thousands) for a house with 0 square feet
- D. The minimum possible house price

6 (3 points)

You fit two regression models: Model A has $R^2 = 0.45$ and Model B has $R^2 = 0.82$. Which statement is correct?

- A. Model B explains more variance in the response variable than Model A
- B. Model B will always make better predictions than Model A
- C. Model A is preferred because lower R^2 means less overfitting
- D. The models must have the same predictors

7 (3 points)

When examining a residual plot (residuals vs. fitted values) for a regression model, which pattern suggests a potential problem?

- A. Residuals randomly scattered around zero with no clear pattern
- B. Residuals forming a fan shape (variance increasing with fitted values)
- C. Most residuals close to zero
- D. Equal numbers of positive and negative residuals

8 (3 points)

Consider the following function:

```
calculate_stats <- function(x) {  
  mean_x <- mean(x)  
  sd_x <- sd(x)  
  mean_x  
}
```

What does this function return?

- A. Both the mean and standard deviation
- B. Only the mean
- C. Only the standard deviation
- D. Nothing (NULL)

9 (3 points)

In logistic regression, what is the response variable?

- A. A continuous numeric outcome
- B. A binary (0/1) outcome
- C. A count variable
- D. A categorical variable with 3 or more levels

10 (3 points)

When fitting a logistic regression model using `glm()`, which argument specifies that the response variable is binary?

- A. `family = gaussian`
- B. `family = binomial`
- C. `family = poisson`
- D. `method = "logistic"`

Part 1B (closed computer): Short answer

Write your answers in complete sentences where appropriate.

11 (5 points)

Inference: Researchers want to test if a new teaching method improves student scores compared to the standard method. They collect data from two independent classes.

The output of a two-sample t-test is shown below:

```
Welch Two Sample t-test

data:  scores_new and scores_standard
t = 2.15, df = 37.8, p-value = 0.019
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 1.243   Inf
sample estimates:
mean of x mean of y
 82.4      74.8
```

State the null and alternative hypotheses for this test. Based on the p-value (using $\alpha = 0.05$), what is your conclusion in the context of the problem?

12 (5 points)

Linear Regression: We fit a simple linear regression model to predict daily temperature (in Fahrenheit) based on wind speed (in mph) using the `airquality` dataset from New York in 1973.

```

Call:
lm(formula = Temp ~ Wind, data = airquality)

Residuals:
    Min       1Q   Median       3Q      Max
-23.291  -5.723   1.709   6.016  19.199

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  90.1349     2.0522  43.921  < 2e-16 ***
Wind        -1.2305     0.1944  -6.331 2.64e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.442 on 151 degrees of freedom
Multiple R-squared:  0.2098,    Adjusted R-squared:  0.2045
F-statistic: 40.08 on 1 and 151 DF,  p-value: 2.642e-09

```

Based on the regression output above, write out the fitted regression equation. Then interpret the slope coefficient in the context of this problem.

13 (5 points)

Prediction: Using the regression model from the previous question, calculate the predicted temperature when the wind speed is 10 mph. Show your work.

14 (5 points)

Functions: Write a simple R function named `is_passing` that takes a single numeric argument `score`.

- If the `score` is greater than or equal to 60, the function should return the character string `"Pass"`.
- Otherwise, it should return `"Fail"`.